

21.2 Detecting heteroscedasticity

The errors are **heteroscedastic** when their variances differ, $\text{Var}(\varepsilon_i) = \sigma_i^2$. Least squares remains unbiased, but its covariance becomes $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ with $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, and $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ no longer estimates it (Theorem 19.2.2). Least squares also stops being the best linear unbiased estimator. How far the usual standard errors go wrong depends on how the variances line up with the regressors: the damage is greatest when the observations with the most influence on a coefficient also have the largest variances. Chapter 15 showed the same thing for the one-way layout, where unequal variances bias the F test only in unbalanced designs (Theorem 15.5.2).

Unequal variances have familiar sources. A response that is an average of m_i observations has variance proportional to $1/m_i$. Counts, sizes and amounts of money tend to vary more when they are larger. And if a coefficient itself varies from unit to unit, $y_i = \mathbf{x}_{(i)}^\top (\boldsymbol{\beta} + \mathbf{u}_i)$ with random $\mathbf{u}_i$, the variance of y_i is a quadratic function of $\mathbf{x}_{(i)}$ (Exercise (B1)).

21.2.1 Residual plots

Under heteroscedasticity the residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ have covariance $(\mathbf{I} - \mathbf{M})\mathbf{\Sigma}(\mathbf{I} - \mathbf{M})$, so

$$E \hat{\varepsilon}_i^2 = (1 - h_{ii})^2 \sigma_i^2 + \sum_{j \neq i} m_{ij}^2 \sigma_j^2. \quad (21.2.1)$$

When h_{ii} is small the first term dominates, and the squared residual tracks the variance of its own observation. At high leverage it does not: a point with h_{ii} near one has a small residual whatever its variance. Even with constant variance the residuals have unequal variances $\sigma^2(1 - h_{ii})$, so plots for heteroscedasticity should use the studentized residuals $r_i = \hat{\varepsilon}_i / (s\sqrt{1 - h_{ii}})$, which have constant variance under the model.

The standard displays are:

- r_i against the fitted values. A fan that opens to the right says that variance grows with the mean.
- r_i against each regressor, and against variables outside the model that might affect precision, such as time, the instrument or the interviewer.
- $|r_i|$, or $\log|r_i|$, against the same variables. Absolute values fold the two halves of the fan onto each other and make a trend in spread easier to see.

The logarithmic plot also estimates a power law. If $\sigma_i = \sigma v_i^\theta$ for a positive variable v_i , then $\varepsilon_i = \sigma v_i^\theta \zeta_i$ with ζ_i of constant law, and $\log|\varepsilon_i| = \log \sigma + \theta \log v_i + \log|\zeta_i|$. So the slope of $\log|r_i|$ on $\log v_i$ estimates θ , with the residuals standing in for the errors.

Example. Engel's food expenditure data

Section 6.9 fitted Engel's data on the annual income and food expenditure of 235 Belgian households, and noted that the spread of expenditure grows with income. The least squares slope is 0.4852. Figure 21.2.1 shows the residual plots. Panel (a) is a fan. Panel (b) plots $|r_i|$ against income on logarithmic scales. The slope of the fitted line, 0.96, suggests a standard deviation roughly proportional to income, the weighting used in Example (Engel's food expenditure data). The most extreme point, at the far right of panel (a), is the household with the largest income. Its leverage is 0.255, about thirty times the average p/n , and its

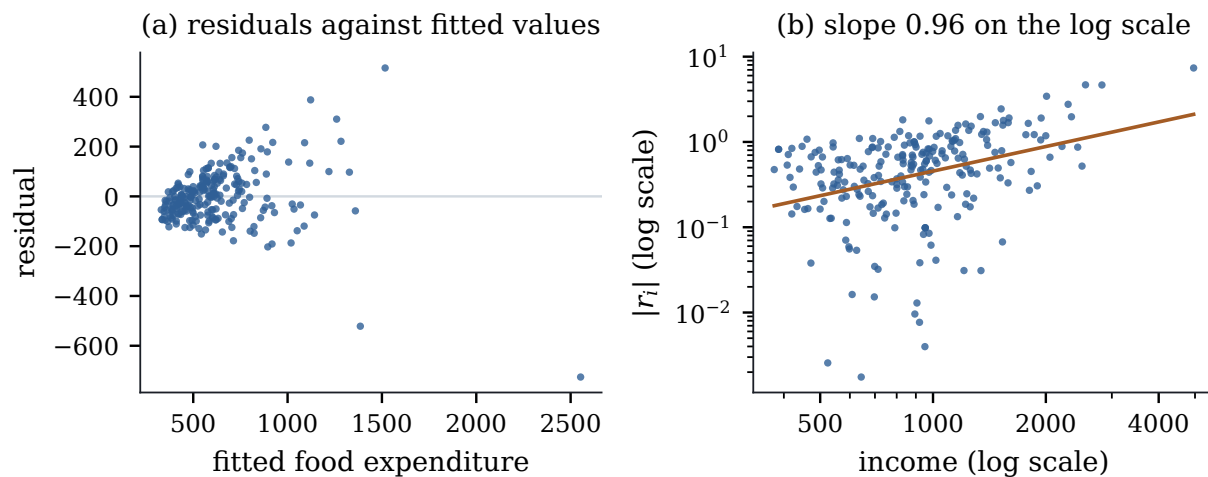


Figure 21.2.1: Residuals from the least squares fit to Engel's data. (a) Residuals against fitted values. (b) Absolute studentized residuals against income, both on logarithmic scales, with the least squares line.

residual is large and negative.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])
beta, *_ = np.linalg.lstsq(X, food, rcond=None)
fitted = X @ beta
e = food - fitted # OLS residuals
Q, _ = np.linalg.qr(X)
h = np.sum(Q**2, axis=1)
s2 = e @ e / (n - 2)
# internally studentized residuals
r = e / np.sqrt(s2 * (1 - h))
```

[Run this code online](#)

Plots show the form of the variance function, which weighting will need. A formal test helps when the pattern is unclear or many models must be screened.

21.2.2 Limit theorems we need

Most results in the rest of the chapter are large-sample results. Three standard limit theorems, the central limit theorem, Slutsky's lemma and the delta method, were stated in [Section 14.4](#), and [Section 19.3](#) recalled three more: the Lindeberg–Feller central limit theorem, the Cramér–Wold device and the continuous mapping theorem. We add two pieces of notation and one law of large numbers.

Remark (Orders in probability and the weak law). A sequence T_n is **bounded in probability**, written $T_n = O_p(1)$, if for every $\eta > 0$ there is a K with $\Pr(|T_n| > K) < \eta$ for all n .

If $E|T_n|$ is bounded, then $T_n = O_p(1)$ by Markov's inequality. We write $T_n = o_p(1)$ if $T_n \rightarrow 0$ in probability. Products and sums obey $O_p(1) o_p(1) = o_p(1)$ and $O_p(1) + o_p(1) = O_p(1)$. The *weak law of large numbers* says that if $W_1, W_2, \dots$ are independent copies of a random variable with finite mean, then $\bar{W}_n \rightarrow E W_1$ in probability. Proofs are in van der Vaart (1998, chapter 2).

The regression version of the central limit theorem concerns a weighted sum $\mathbf{c}^\top \boldsymbol{\xi}$ of independent variables, with weights determined by the design. It holds as long as no single weight dominates. Lemma 19.3.1 proved it for identically distributed variables. The sandwich estimators of Section 21.4 also need it for variables whose variances differ, and for that a moment slightly higher than the second must be bounded, in the form due to Lyapunov.

Lemma 21.2.1. A central limit theorem for weighted sums

For each n let $\xi_1, \dots, \xi_n$ be independent with mean zero, and let $\mathbf{c} = \mathbf{c}_n \in \mathbb{R}^n$ be nonzero weights with

$$\max_i \frac{c_i^2}{\|\mathbf{c}\|^2} \rightarrow 0. \quad (21.2.2)$$

Suppose either (a) the ξ_i have a common distribution, not depending on n , with variance $\tau^2 \in (0, \infty)$, or (b) for some constants $\delta, K, \tau_{\min} > 0$, $\text{Var } \xi_i \geq \tau_{\min}^2$ and $E|\xi_i|^{2+\delta} \leq K$ for all i and n . Then

$$\frac{\sum_i c_i \xi_i}{(\sum_i c_i^2 \text{Var } \xi_i)^{1/2}} \rightarrow N(0, 1) \quad \text{in distribution.}$$

Proof. Case (a) is the lemma on weighted sums of errors (Lemma 19.3.1) in Section 19.3, applied with ξ_i in place of the errors and the unit weights $\mathbf{c}/\|\mathbf{c}\|$. For case (b), let $v_n = \sum_i c_i^2 \text{Var } \xi_i$ and $X_{ni} = c_i \xi_i / \sqrt{v_n}$, so that $\sum_i \text{Var } X_{ni} = 1$, and write $\mu_n = \max_i |c_i|/\|\mathbf{c}\| \rightarrow 0$. We check Lindeberg's condition. Here $v_n \geq \tau_{\min}^2 \|\mathbf{c}\|^2$, and on $\{|X_{ni}| > \eta\}$ we have $X_{ni}^2 \leq |X_{ni}|^{2+\delta}/\eta^\delta$. Hence the Lindeberg sum satisfies

$$\begin{aligned} \sum_i E[X_{ni}^2 \mathbf{1}\{|X_{ni}| > \eta\}] &\leq \frac{K \sum_i |c_i|^{2+\delta}}{\eta^\delta v_n^{1+\delta/2}} \\ &\leq \frac{K \mu_n^\delta \|\mathbf{c}\|^2 \|\mathbf{c}\|^\delta}{\eta^\delta \tau_{\min}^{2+\delta} \|\mathbf{c}\|^{2+\delta}} = \frac{K \mu_n^\delta}{\eta^\delta \tau_{\min}^{2+\delta}} \rightarrow 0, \end{aligned}$$

using $\sum_i |c_i|^{2+\delta} \leq \max_i |c_i|^\delta \sum_i c_i^2$. The Lindeberg–Feller theorem gives the conclusion. ■

For the coefficient vector $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ of a linear function $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$, the condition Eq. 21.2.2 is implied by $\max_i h_{ii} \rightarrow 0$, since $c_i^2 \leq h_{ii} \|\mathbf{c}\|^2$, as the proof of Theorem 19.3.2 shows. This is the condition behind that theorem.

21.2.3 The Breusch–Pagan score test

To test for heteroscedasticity we need an alternative. A **variance function** model supposes that

$$\sigma_i^2 = h(\alpha_0 + \mathbf{z}_i^\top \boldsymbol{\alpha}), \quad (21.2.3)$$

where $\mathbf{z}_i \in \mathbb{R}^q$ is a vector of observed variables, possibly some or all of the regressors, and h is a positive function. Constant variance is the hypothesis $\boldsymbol{\alpha} = \mathbf{0}$. Common choices are $h(t) = e^t$, which keeps variances positive, and $h(t) = t$. The score test of Section 11.4 needs only the fit under the hypothesis, which is ordinary least squares, and it turns out not to depend on h at all.

Write $\tilde{\sigma}^2 = \|\hat{\varepsilon}\|^2/n$ for the maximum likelihood estimate of σ^2 under the hypothesis, $u_i = \hat{\varepsilon}_i^2/\tilde{\sigma}^2$ for the scaled squared residuals, and $\tilde{\mathbf{Z}}$ for the $n \times q$ matrix whose rows are $(\mathbf{z}_i - \bar{\mathbf{z}})^\top$. Let R^2 be the coefficient of determination from regressing $\hat{\varepsilon}_i^2$ on an intercept and $\mathbf{z}_i$, and let $\hat{\kappa} = n \sum_i \hat{\varepsilon}_i^4 / (\sum_i \hat{\varepsilon}_i^2)^2$ be the sample kurtosis of the residuals. In this section κ denotes the raw kurtosis μ_4/σ^4 , which equals 3 for normal errors; Section 19.3 used the same letter for the excess kurtosis $\mu_4/\sigma^4 - 3$.

Theorem 21.2.2. Breusch–Pagan and Koenker

Let $\tilde{\mathbf{Z}}$ have full column rank q .

- (a) (*Score statistic.*) Suppose $\varepsilon_i \sim N(0, \sigma_i^2)$ independently, with Eq. 21.2.3, h continuously differentiable, and $h' \neq 0$ at the null value of α_0 . The score statistic for $\boldsymbol{\alpha} = \mathbf{0}$ is

$$\text{BP} = \frac{1}{2}(\mathbf{u} - \mathbf{1})^\top \tilde{\mathbf{Z}}(\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}})^{-1} \tilde{\mathbf{Z}}^\top (\mathbf{u} - \mathbf{1}),$$

half the explained sum of squares from regressing u_i on an intercept and $\mathbf{z}_i$. It does not depend on h .

- (b) (*Studentized form.*) Let $K = nR^2$. Then $\text{BP} = \frac{1}{2}(\hat{\kappa} - 1)K$.
- (c) (*Null distributions.*) Suppose instead only that the errors are independent and identically distributed with mean 0, variance σ^2 , finite fourth moment μ_4 and kurtosis $\kappa = \mu_4/\sigma^4 > 1$, that $\text{rank}(\mathbf{X}) = p$ and q are fixed, that the $\mathbf{z}_i$ are bounded, and that $n^{-1}\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} \rightarrow \mathbf{Q}_z$, positive definite. Then, as $n \rightarrow \infty$,

$$K \rightarrow \chi^2(q), \quad \text{BP} \rightarrow \frac{1}{2}(\kappa - 1)\chi^2(q) \quad \text{in distribution.}$$

Proof.

- (a) The log-likelihood is $\ell = -\frac{1}{2} \sum_i \log(2\pi\sigma_i^2) - \frac{1}{2} \sum_i (y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})^2/\sigma_i^2$. Put $\boldsymbol{\theta} = (\alpha_0, \boldsymbol{\alpha}^\top)^\top$, $\mathbf{z}_i^* = (1, \mathbf{z}_i^\top)^\top$ and $\eta_i = \alpha_0 + \mathbf{z}_i^\top \boldsymbol{\alpha}$. Since $\partial\ell/\partial\sigma_i^2 = (\varepsilon_i^2/\sigma_i^2 - 1)/(2\sigma_i^2)$ and $\partial\sigma_i^2/\partial\boldsymbol{\theta} = h'(\eta_i)\mathbf{z}_i^*$,

$$\begin{aligned} \frac{\partial\ell}{\partial\boldsymbol{\theta}} &= \sum_i \frac{h'(\eta_i)}{2\sigma_i^2} \left(\frac{\varepsilon_i^2}{\sigma_i^2} - 1 \right) \mathbf{z}_i^*, \\ \frac{\partial\ell}{\partial\boldsymbol{\beta}} &= \sum_i \frac{\varepsilon_i}{\sigma_i^2} \mathbf{x}_{(i)}. \end{aligned}$$

Under normality $\varepsilon_i^2/\sigma_i^2$ has variance 2, and $E\varepsilon_i(\varepsilon_i^2 - \sigma_i^2) = 0$, so the information matrix

is block diagonal between $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$, and its $\boldsymbol{\theta}$ block is $\sum_i h'(\eta_i)^2 \mathbf{z}_i^* \mathbf{z}_i^{*\top} / (2\sigma_i^4)$. Under the hypothesis $\eta_i = \alpha_0$ for all i , the restricted maximum likelihood estimates are $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$, and $h(\hat{\alpha}_0) = \hat{\sigma}^2$. Write $g = h'(\hat{\alpha}_0)$, let $\mathbf{Z}$ be the $n \times q$ matrix with rows $\mathbf{z}_i^\top$, and put $\mathbf{Z}^* = [\mathbf{1}, \mathbf{Z}]$. At the restricted estimate the $\boldsymbol{\beta}$ score vanishes, and

$$\mathbf{S}_\theta = \frac{g}{2\hat{\sigma}^2} \mathbf{Z}^{*\top} (\mathbf{u} - \mathbf{1}), \quad \mathcal{I}_{\theta\theta} = \frac{g^2}{2\hat{\sigma}^4} \mathbf{Z}^{*\top} \mathbf{Z}^*.$$

By block diagonality the score statistic is $\mathbf{S}_\theta^\top \mathcal{I}_{\theta\theta}^{-1} \mathbf{S}_\theta = \frac{1}{2} (\mathbf{u} - \mathbf{1})^\top \mathbf{Z}^* (\mathbf{Z}^{*\top} \mathbf{Z}^*)^{-1} \mathbf{Z}^{*\top} (\mathbf{u} - \mathbf{1})$, in which g has cancelled. The matrix here is the projection onto $\mathcal{C}(\mathbf{Z}^*) = \mathcal{C}(\mathbf{1}) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$, and $\mathbf{1}^\top (\mathbf{u} - \mathbf{1}) = 0$ because $\sum_i u_i = n$, so only the projection onto $\mathcal{C}(\tilde{\mathbf{Z}})$ contributes (Lemma 6.6.1).

- (b) Let $\mathbf{w}$ have entries $\hat{\varepsilon}_i^2$. Its explained sum of squares on $[\mathbf{1}, \mathbf{Z}]$ is $\text{ESS}_w = (\mathbf{w} - \hat{\sigma}^2 \mathbf{1})^\top \tilde{\mathbf{M}} (\mathbf{w} - \hat{\sigma}^2 \mathbf{1})$, where $\tilde{\mathbf{M}}$ projects onto $\mathcal{C}(\tilde{\mathbf{Z}})$, and its total sum of squares is $\text{TSS}_w = \sum_i (\hat{\varepsilon}_i^2 - \hat{\sigma}^2)^2 = \sum_i \hat{\varepsilon}_i^4 - n\hat{\sigma}^4$. Since $\mathbf{u} - \mathbf{1} = (\mathbf{w} - \hat{\sigma}^2 \mathbf{1}) / \hat{\sigma}^2$, (a) gives $\text{BP} = \text{ESS}_w / (2\hat{\sigma}^4)$, while $K = n \text{ESS}_w / \text{TSS}_w$. Their ratio is $\text{TSS}_w / (2n\hat{\sigma}^4) = (\hat{\kappa} - 1)/2$.
- (c) Put $\mathbf{a} = \mathbf{M}\boldsymbol{\varepsilon}$, so that $\hat{\varepsilon}_i = \varepsilon_i - a_i$ and $\mathbb{E}\|\mathbf{a}\|^2 = \sigma^2 p$ (Proposition 21.1.1). Let $|\tilde{z}_{ik}| \leq C$. By (b),

$$K = \frac{\mathbf{T}_n^\top (n^{-1} \tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}})^{-1} \mathbf{T}_n}{\text{TSS}_w / n},$$

$$\mathbf{T}_n = n^{-1/2} \tilde{\mathbf{Z}}^\top \mathbf{w} = n^{-1/2} \tilde{\mathbf{Z}}^\top (\mathbf{w} - \sigma^2 \mathbf{1}),$$

using $\tilde{\mathbf{Z}}^\top \mathbf{1} = \mathbf{0}$. We show three things.

The leading term. Let $\mathbf{T}_n^0 = n^{-1/2} \sum_i \tilde{\mathbf{z}}_i (\varepsilon_i^2 - \sigma^2)$. The $\xi_i = \varepsilon_i^2 - \sigma^2$ are independent and identically distributed with variance $\mu_4 - \sigma^4 > 0$. For fixed $\mathbf{b} \in \mathbb{R}^q$ the weights $c_i = \mathbf{b}^\top \tilde{\mathbf{z}}_i$ are bounded and $\|\mathbf{c}\|^2 / n \rightarrow \mathbf{b}^\top \mathbf{Q}_z \mathbf{b} > 0$, so Eq. 21.2.2 holds, and Lemma 21.2.1(a) shows that $\mathbf{b}^\top \mathbf{T}_n^0 \rightarrow N(0, (\mu_4 - \sigma^4) \mathbf{b}^\top \mathbf{Q}_z \mathbf{b})$. By the Cramér–Wold device, $\mathbf{T}_n^0 \rightarrow N_q(\mathbf{0}, (\mu_4 - \sigma^4) \mathbf{Q}_z)$.

The residuals do not matter. $\hat{\varepsilon}_i^2 - \varepsilon_i^2 = a_i^2 - 2a_i \varepsilon_i$, so the k th coordinate of $\mathbf{T}_n - \mathbf{T}_n^0$ is $n^{-1/2} \sum_i \tilde{z}_{ik} a_i^2 - 2n^{-1/2} \boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}$, with $\mathbf{B} = \frac{1}{2}(\mathbf{D}\mathbf{M} + \mathbf{M}\mathbf{D})$ and $\mathbf{D} = \text{diag}(\tilde{z}_{1k}, \dots, \tilde{z}_{nk})$. The first term is at most $Cn^{-1/2} \|\mathbf{a}\|^2$, whose mean $C\sigma^2 p / \sqrt{n}$ tends to zero. For the second, $|\mathbb{E} \boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}| = \sigma^2 |\text{tr}(\mathbf{D}\mathbf{M})| \leq C\sigma^2 p$ (Theorem 2.3.1), and by Theorem 2.3.3

$$\begin{aligned} \text{Var}(\boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}) &\leq |\mu_4 - 3\sigma^4| \sum_i b_{ii}^2 + 2\sigma^4 \text{tr}(\mathbf{B}^2) \\ &\leq (|\mu_4 - 3\sigma^4| + 2\sigma^4) C^2 p, \end{aligned}$$

because $b_{ii} = \tilde{z}_{ik} h_{ii}$, $\sum_i h_{ii}^2 \leq \sum_i h_{ii} = p$, and $\text{tr}(\mathbf{B}^2) \leq \|\mathbf{D}\mathbf{M}\|_F^2 = \text{tr}(\mathbf{M}\mathbf{D}^2\mathbf{M}) \leq C^2 \text{tr} \mathbf{M}$. So $n^{-1/2} \boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}$ has mean and variance tending to zero, and $\mathbf{T}_n - \mathbf{T}_n^0 = o_p(1)$.

The denominator. $\hat{\sigma}^2 = (\|\boldsymbol{\varepsilon}\|^2 - \|\mathbf{a}\|^2) / n \rightarrow \sigma^2$ by the weak law and $\|\mathbf{a}\|^2 = O_p(1)$. For the fourth powers, Minkowski's inequality in the norm $\|\mathbf{v}\|_4 = (\sum_i v_i^4)^{1/4}$ gives $|\|\hat{\boldsymbol{\varepsilon}}\|_4 - \|\boldsymbol{\varepsilon}\|_4| \leq \|\mathbf{a}\|_4 \leq \|\mathbf{a}\| = O_p(1)$, while $n^{-1} \|\boldsymbol{\varepsilon}\|_4^4 \rightarrow \mu_4$ by the weak law. Hence $n^{-1} \sum_i \hat{\varepsilon}_i^4 \rightarrow \mu_4$, $\text{TSS}_w / n \rightarrow \mu_4 - \sigma^4$ and $\hat{\kappa} \rightarrow \kappa$.

Combining these by Slutsky's lemma and continuous mapping, K converges in distribution to $\mathbf{T}^\top \mathbf{Q}_z^{-1} \mathbf{T} / (\mu_4 - \sigma^4)$ with $\mathbf{T} \sim N_q(\mathbf{0}, (\mu_4 - \sigma^4) \mathbf{Q}_z)$, which is $\chi^2(q)$ by Corollary 4.3.3. The statement about BP follows from (b). ■

Part (a) is the test of Breusch and Pagan (1979). Cook and Weisberg (1983) derived the same

statistic, and used the regression of u_i on $\mathbf{z}_i$ as a diagnostic plot. Part (c) exposes a weakness. BP is calibrated for normal errors, which have $\kappa = 3$. With heavier tails its null distribution is inflated by the factor $(\kappa - 1)/2$, and the test rejects too often, detecting kurtosis rather than heteroscedasticity. Koenker (1981) proposed the studentized form nR^2 , whose null limit is the same for every error law with a fourth moment. It is the version to use.

Example. Size of the two tests

Take a straight-line regression with $n = 100$ equally spaced values of x , constant variance, and $\mathbf{z}_i = x_i$. In 40,000 simulated data sets the 5% tests rejected as follows:

error law	κ	Breusch–Pagan	its limit	Koenker
normal	3	0.048	0.050	0.050
Laplace	6	0.185	0.215	0.047
uniform	1.8	0.003	0.002	0.049

The Laplace law makes the Breusch–Pagan test reject nearly four times too often; the uniform law makes it almost never reject. The studentized test holds its level in all three cases.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(2105)
n = 100
x = np.linspace(0, 10, n)
X = np.column_stack([np.ones(n), x])
Q, _ = np.linalg.qr(X)
# centred, unit-length variance regressor
zc = (x - x.mean()) / np.linalg.norm(x - x.mean())
crit = stats.chi2.ppf(0.95, 1)
laws = {
    "normal": (lambda size: rng.normal(size=size), 3.0),
    "Laplace": (lambda size: rng.laplace(scale=np.sqrt(0.5), size=size), 6.0),
    "uniform": (lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size=size),
               1.8),
}

def size_table(reps):
    table = {}
    for name, (draw, kappa) in laws.items():
        E = draw((reps, n))
        R = E - (E @ Q) @ Q.T # residuals
        R2 = R**2
        sig2 = R2.mean(axis=1, keepdims=True)
        # explained SS of  $e^2$  on  $[1, x]$ 
        ess = (R2 @ zc) ** 2
        bp = ess / (2 * sig2[:, 0] ** 2)
        koenker = n * ess / np.sum((R2 - sig2) ** 2, axis=1)
        limit_bp = stats.chi2.sf(crit / ((kappa - 1) / 2), 1)
        table[name] = (np.mean(bp > crit), np.mean(koenker > crit), limit_bp)
    print(f"{name:8s}: Breusch-Pagan rejects {table[name][0]:.3f} "
          f"(limit {limit_bp:.3f}), "
          f"Koenker rejects {table[name][1]:.3f}")
```

```

return table

# a quick version (the book uses 40000)
quick = size_table(2000)

```

[Run this code online](#)

White's test. For the least squares standard errors, what matters is whether σ_i^2 is correlated with the entries of $\mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top$, the terms of $\mathbf{X}^\top\boldsymbol{\Sigma}\mathbf{X}$ (White 1980). White's test is Koenker's statistic with $\mathbf{z}_i$ the regressors, their squares and their cross products; with many regressors it has many degrees of freedom and low power.

Example. Testing Engel's data

With $\mathbf{z}_i = \text{income}_i$, the Breusch–Pagan statistic is 636.0 and Koenker's is 109.3, on one degree of freedom. Their ratio is $5.820 = (\hat{\kappa} - 1)/2$, since the residuals have sample kurtosis 12.64. The heavy tails of the residuals inflate BP about sixfold, but the evidence is overwhelming either way: Koenker's p-value is 1.4×10^{-25} . White's test, with income and its square, gives 181.1 on two degrees of freedom.

```

def explained_ss(v, Z):
    """Explained sum of squares of v on [1, Z], about the mean of v."""
    Z1 = np.column_stack([np.ones(len(v)), Z])
    coef, *_ = np.linalg.lstsq(Z1, v, rcond=None)
    return np.sum((Z1 @ coef - v.mean()) ** 2)

# maximum likelihood estimate under H0
sig2 = e @ e / n
u = e**2 / sig2
# Breusch-Pagan score statistic
bp = explained_ss(u, income) / 2
tss = np.sum((e**2 - sig2) ** 2)
# n R^2 from regressing e^2 on [1, income]
koenker = n * explained_ss(e**2, income) / tss
white = n * explained_ss(e**2, np.column_stack([income, income**2])) / tss
# sample kurtosis of the residuals
kappa = n * np.sum(e**4) / np.sum(e**2) ** 2
print(f"Breusch-Pagan {bp:.1f}, Koenker {koenker:.1f}, "
      f"White {white:.1f} (2 df)")
print(f"ratio BP/Koenker = {bp / koenker:.3f}, "
      f"(kurtosis - 1)/2 = {(kappa - 1) / 2:.3f}")

# Goldfeld-Quandt: sort, drop the middle fifth
order = np.argsort(income)
m = (n - 47) // 2
low, high = order[:m], order[-m:]
sse = []
for g in (low, high):
    b_g = np.linalg.lstsq(X[g], food[g], rcond=None)[0]
    sse.append(np.sum((food[g] - X[g] @ b_g) ** 2))
gq = (sse[1] / (m - 2)) / (sse[0] / (m - 2))
print(f"Goldfeld-Quandt F = {gq:.2f} on ({m - 2}, {m - 2}) df, "
      f"p = {stats.f.sf(gq, m - 2, m - 2):.1e}")

```

[Run this code online](#)

21.2.4 An exact test: Goldfeld–Quandt

When the variance is suspected to change monotonically with one variable v that does not depend on the responses, there is an exact test under normality. Sort the observations by v , set aside a block in the middle, and fit the model separately to the n_1 observations with the smallest values of v and the n_2 with the largest.

Proposition 21.2.3. Goldfeld–Quandt

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, let the two groups be chosen without reference to $\mathbf{Y}$, and let $\mathbf{X}_1$ and $\mathbf{X}_2$, their rows of $\mathbf{X}$, have rank $p < n_k$. If SSE_1 and SSE_2 are the residual sums of squares of the separate fits, then

$$F_{\text{GQ}} = \frac{\text{SSE}_2/(n_2 - p)}{\text{SSE}_1/(n_1 - p)} \sim F(n_2 - p, n_1 - p).$$

Proof. Each group is a normal linear model with error variance σ^2 , so $\text{SSE}_k/\sigma^2 \sim \chi^2(n_k - p)$ by Theorem 7.5.1(b). The two sums of squares are functions of disjoint sets of independent errors, so they are independent, and the ratio has the stated law by Definition 4.1.2 with $\gamma = 0$. ■

Dropping the middle block sharpens the contrast between the groups at the cost of degrees of freedom, and how much to drop is a matter of judgement; here we omit a fifth of the data. For Engel's data, with the middle 47 households omitted and 94 in each group, $F_{\text{GQ}} = 10.22$ on $(92, 92)$ degrees of freedom, with p-value 1.7×10^{-24} . The test is exact only under normality. Like every comparison of variances, it is sensitive to the tails of the error law. When the observations fall into groups, a more robust comparison is Levene's test (Levene 1960), the one-way analysis of variance of the absolute residuals $|\hat{\varepsilon}_i|$, or its version with deviations from group medians (Brown and Forsythe 1974). Box (1953) showed how badly the classical normal-theory test of equal group variances, Bartlett's, is affected by kurtosis.

21.2.5 Exercises

21.2.5.1 A. Check your understanding

Exercise. A1

Derive Eq. 21.2.1. Show that under constant variance $E\hat{\varepsilon}_i^2 = \sigma^2(1 - h_{ii})$, and give an example in which σ_i^2 is the largest variance but $E\hat{\varepsilon}_i^2$ is the smallest.

21.2.5.2 B. Practice

Exercise. B1

Let $y_i = \mathbf{x}_{(i)}^\top(\boldsymbol{\beta} + \mathbf{u}_i) + \varepsilon_i$, where $\mathbf{u}_i$ has mean $\mathbf{0}$ and covariance $\boldsymbol{\Omega}$, ε_i has variance σ^2 , and

all are independent. Show that $\text{Var}(y_i)$ is a linear function of the squares and cross products of the regressors. Which test of this section is the score-type test for $\boldsymbol{\Omega} = \mathbf{0}$?

Solution. $y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta} = \mathbf{x}_{(i)}^\top \mathbf{u}_i + \varepsilon_i$ has variance $\sigma^2 + \mathbf{x}_{(i)}^\top \boldsymbol{\Omega} \mathbf{x}_{(i)} = \sigma^2 + \sum_{j,k} \omega_{jk} x_{ij} x_{ik}$, linear in the products $x_{ij} x_{ik}$ with coefficients ω_{jk} . This is Eq. 21.2.3 with $h(t) = t$ and $\mathbf{z}_i$ the distinct squares and cross products (the squares of the intercept column merge with the intercept). The score test is the Breusch–Pagan test with these $\mathbf{z}_i$, whose studentized form is White’s test.

Exercise. B2

Under the assumptions of Proposition 21.2.3, suppose the observations are sorted by their fitted values $\hat{Y}_i$ instead of by a fixed variable. Show that F_{GQ} still has the $F(n_2 - p, n_1 - p)$ distribution. Show by an example that sorting by $|\hat{\varepsilon}_i|$ destroys the result.

Solution. Fix a partition, with selection matrices $\mathbf{S}_1, \mathbf{S}_2$, and let $\mathbf{M}_k$ project onto $\mathcal{C}(\mathbf{X}_k)$. The group residuals $(\mathbf{I} - \mathbf{M}_k) \mathbf{S}_k \boldsymbol{\varepsilon}$ and the fitted values $\mathbf{M} \boldsymbol{\varepsilon}$ are jointly normal with cross-covariance $\sigma^2 (\mathbf{I} - \mathbf{M}_k) \mathbf{S}_k \mathbf{M} = \sigma^2 (\mathbf{I} - \mathbf{M}_k) \mathbf{X}_k (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{0}$, since $\mathbf{S}_k \mathbf{X} = \mathbf{X}_k$. So, for every fixed partition, $(\text{SSE}_1, \text{SSE}_2)$ is independent of $\mathbf{M} \mathbf{Y}$, and it has the law of Proposition 21.2.3. Sorting by fitted values makes the partition a function of $\mathbf{M} \mathbf{Y}$, so given $\mathbf{M} \mathbf{Y}$ the statistic has the F law, and therefore also unconditionally. Sorting by $|\hat{\varepsilon}_i|$ puts the largest residuals in the second group: with $p = 1$, $\mathbf{X} = \mathbf{1}$ and $n = 4$, no middle block, the second group contains the two observations farthest from $\bar{y}$, and F_{GQ} exceeds one with probability one.

21.2.5.3 C. Going deeper

Exercise. C1

Suppose the observations fall into g groups and $\mathbf{z}_i$ consists of $g - 1$ group indicators. Show that Koenker’s statistic is n times the R^2 of a one-way analysis of variance of the squared residuals, and compare it with Levene’s test.